

Q 1

2001

Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$.

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x+2)} = \frac{12}{4} = 3$$

Q 2

2002

Differentiate $x^3 + 2$.

$$\frac{d}{dx}(x^3 + 2) = 3x^2.$$

Q 3

2003

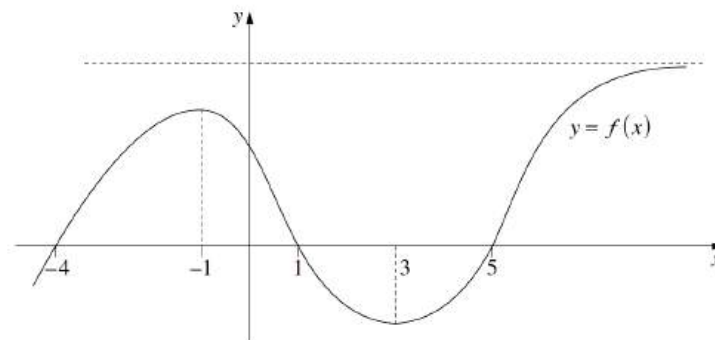
Differentiate $x^4 + 5x^{-1}$ with respect to x .

$$\frac{d}{dx}(x^4 + 5x^{-1}) = 4x^3 - 5x^{-2}.$$

Q 4

2009

(a)



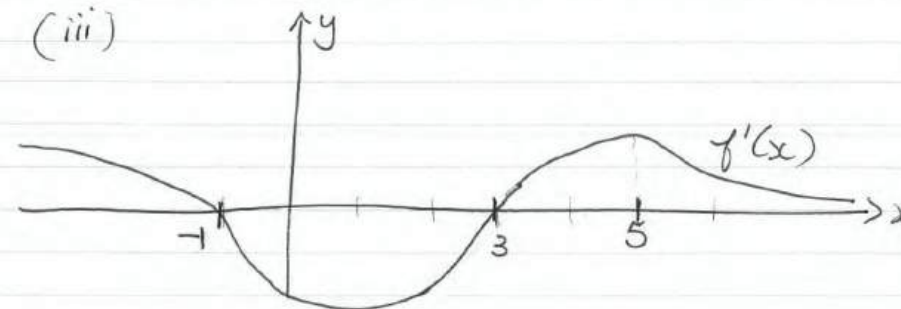
The diagram shows the graph of a function $y = f(x)$.

- (i) For which values of x is the derivative, $f'(x)$, negative? 1
- (ii) What happens to $f'(x)$ for large values of x ? 1
- (iii) Sketch the graph $y = f'(x)$. 2

(a) (i) $f'(x) < 0$ for $-1 < x < 3$

(ii) $f'(x) \rightarrow 0$ as $x \rightarrow \infty$

(iii)

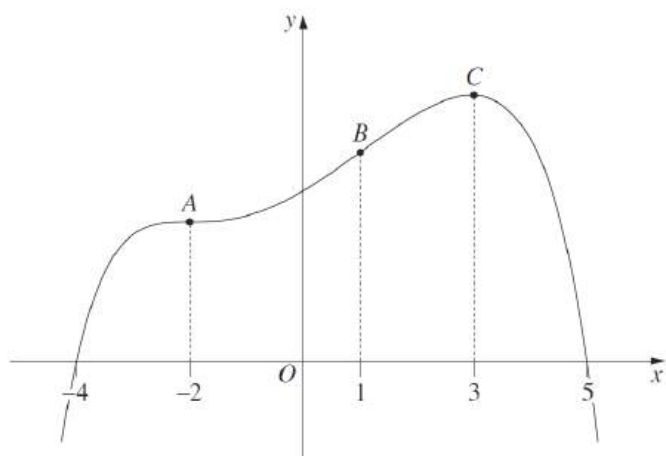


Q 5

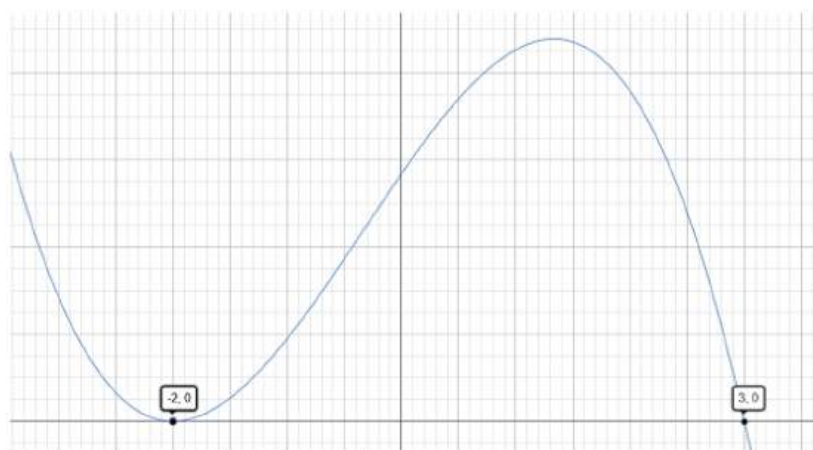
2014

The diagram shows the graph of a function $f(x)$.

The graph has a horizontal point of inflexion at A , a point of inflexion at B and a maximum turning point at C .



Sketch the graph of the derivative $f'(x)$.



Note a maximum turning point at $x = 1$.

Q 1
2001

Find the equation of the tangent to the curve $y = x^2 + 3x$ at the point $(1, 4)$.

$$y = x^2 + 3x$$

$$\frac{dy}{dx} = 2x + 3.$$

$$\text{At } x = 1, \frac{dy}{dx} = 5,$$

$$\therefore \text{gradient of tangent} = 5.$$

$$\begin{aligned} \text{The tangent is } y - 4 &= 5(x - 1) \\ y &= 5x - 1. \end{aligned}$$

Q 2
2009

Find the gradient of the tangent to the curve $y = x^4 - 3x$ at the point $(1, -2)$.

$$\begin{aligned} y &= x^4 - 3x \\ \frac{dy}{dx} &= 4x^3 - 3 \\ \text{When } x &= 1, \frac{dy}{dx} = 4 - 3 = 1 \end{aligned}$$

Q 3
2011

Find the equation of the tangent to the curve $y = (2x + 1)^4$ at the point where $x = -1$.

$$\begin{aligned} y &= (2x + 1)^4 \\ y' &= 4 \times (2x + 1)^3 \times 2 \\ &= 8(2x + 1)^3 \end{aligned}$$

$$\begin{aligned} \text{When } x &= -1, y' = 8(-2 + 1)^3 \\ &= -8 \end{aligned}$$

$$\begin{aligned} \therefore y - y_1 &= m(x - x_1) \\ y - 1 &= -8(x - (-1)) \\ y - 1 &= -8x - 8 \\ 8x + y + 7 &= 0 \end{aligned}$$

Q 4
2012

Find the equation of the tangent to the curve $y = x^2$ at the point where $x = 3$.

$$y = x^2, \quad \frac{dy}{dx} = 2x$$

$$\text{slope of tangent at } x = 3 \text{ is } 2 \times 3 = 6$$

$$\therefore 6 = \frac{y - 3^2}{x - 3} = \frac{y - 9}{x - 3}$$

$$\begin{aligned} \text{Hence the equation of the tangent is } y - 9 &= 6(x - 3) + 9 \\ &= 6x - 9 \end{aligned}$$



INTRODUCTION TO CALCULUS BES

PART D: TANGENTS AND NORMALS

2unit

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Q 5

2017

Find the equation of the tangent to the curve $y = x^2 + 4x - 7$ at the point $(1, -2)$.

$$y = x^2 + 4x - 7$$

$$y' = 2x + 4$$

$$\text{when } x = 1 \quad y' = 2 + 4 = 6$$

$$\therefore m = 6$$

$$y - y_1 = m(x - x_1)$$

$$y - -2 = 6(x - +1)$$

$$y = 6x - 8$$



Q 1
2008

 Differentiate with respect to x : $(x^2 + 3)^9$

$$\frac{d}{dx}(x^2 + 3)^9 = 9(x^2 + 3)^8 (2x) = 18x(x^2 + 3)^8$$

Q 2
2005

 Differentiate with respect to x : $\frac{x^2}{x-1}$

Quotient Rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$

$$\begin{aligned} \frac{d}{dx}\left(\frac{x^2}{x-1}\right) &= \frac{(x-1) \times 2x - x^2 \times 1}{(x-1)^2} \\ &= \frac{2x^2 - 2x - x^2}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2} \end{aligned}$$

Q 3
2014

 Differentiate $\frac{x^3}{x+1}$.

$$\begin{aligned} \frac{d}{dx}\left(\frac{x^3}{x+1}\right) &= \frac{(x+1) \cdot 3x^2 - x^3}{(x+1)^2} \\ &= \frac{x^2(2x+3)}{(x+1)^2} \end{aligned}$$

Q 4
2015

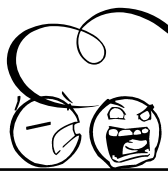
 Find $f'(x)$, where $f(x) = \frac{x^2 + 3}{x-1}$.

$$\begin{aligned} f(x) &= \frac{x^2 + 3}{x-1} \\ f'(x) &= \frac{2x(x-1) - (x^2 + 3)}{(x-1)^2} \\ &= \frac{x^2 - 2x - 3}{(x-1)^2} \end{aligned}$$

Q 5
2016

 Differentiate $\frac{x+2}{3x-4}$.

$$\begin{aligned} \frac{d}{dx}\left(\frac{x+2}{3x-4}\right) &= \frac{(3x-4)(1) - (x+2)(3)}{(3x-4)^2} \\ &= \frac{3x-4-3x-6}{(3x-4)^2} \\ &= \frac{-10}{(3x-4)^2} \end{aligned}$$



INTRODUCTION TO CALCULUS BOS

Part h: Rates of Change



Unit

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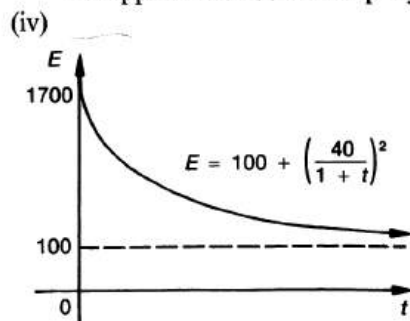
Q 1

1989

The rate of emission E , in tonnes per year, of chloro-fluorocarbons (CFCs) in Australia from 31 October 1989 will be given by

$$E = 100 + \left(\frac{40}{1+t}\right)^2, \text{ where } t \text{ is the time in years.}$$

- (i) What is the rate of emission E on 31 October 1989?
 - (ii) What is the rate of emission E on 31 October 1998?
 - (iii) What value does E approach as the years pass by?
 - (iv) Draw a sketch of E as a function of t .
- (i) On 31 October 1989, $t = 0$
 $\therefore E = 100 + \left(\frac{40}{1}\right)^2$
 $= 1700$ tonnes per year.
- (ii) On 31 October 1998, $t = 9$
 $\therefore E = 100 + \left(\frac{40}{10}\right)^2$
 $= 100 + 16$
 $= 116$ tonnes per year.
- (iii) As t becomes larger, $\left(\frac{40}{1+t}\right)^2$ becomes smaller and approaches zero.
 $\therefore E$ approaches 100 tonnes per year.



Q 2

1990

It is assumed that the number $N(t)$ of termites in a certain mound at time $t \geq 0$ is given by

$$N(t) = \frac{A}{2 + e^{-t}}, \text{ where } A \text{ is a constant, and } t \text{ is measured in months.}$$

- (i) At time $t = 0$, $N(t)$ is estimated at 3×10^5 termites. What is the value of A ?
 - (ii) What is the value of $N(t)$ after one month?
 - (iii) How many termites would you expect to find in the mound when t is very large?
 - (iv) Find an expression for the rate at which the number of termites increases at any time t .
- (i) $N(t) = \frac{A}{2 + e^{-t}}$
 When $t = 0$, $N(t) = 3 \times 10^5$
 $\therefore 3 \times 10^5 = \frac{A}{2 + e^0}$
 $3 \times 10^5 = \frac{A}{3}$
 $\therefore A = 9 \times 10^5$.
- (ii) $N(t) = \frac{9 \times 10^5}{2 + e^{-t}}$
 $N(1) = \frac{9 \times 10^5}{2 + e^{-1}}$
 $= 380\,086.91 \dots$ (by calc.)
 $= 380\,087$ termites.
- (iii) As t becomes very large, e^{-t} becomes very small and approaches zero and hence $N(t)$ approaches
 $\frac{9 \times 10^5}{2} = 4.5 \times 10^5$ termites.
- (iv) $N(t) = 9 \times 10^5 (2 + e^{-t})^{-1}$
 $N'(t) = -9 \times 10^5 (2 + e^{-t})^{-2} \cdot \frac{d}{dt} (2 + e^{-t})$
 $= \frac{-9 \times 10^5}{(2 + e^{-t})^2} \cdot -e^{-t}$
 $= \frac{9 \times 10^5}{e^t (2 + e^{-t})^2}$



INTRODUCTION TO CALCULUS BOS

PART h: Rates of Change



Unit

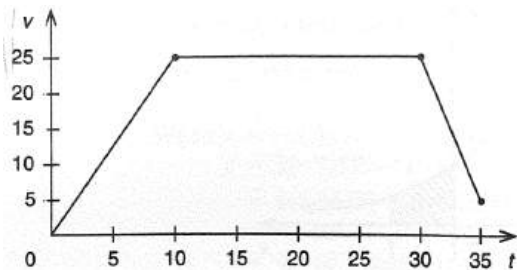
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Q 3

1992

'The car moved away from where it had stopped, its speed increasing at a constant rate, and after exactly 10 seconds it was travelling at 25 m/s. It continued at a constant speed for a further 20 seconds. Then the brakes were applied causing it to slow down at a constant rate, so that 5 seconds later it was travelling at 5 m/s.'

- (i) Let the car's speed be v m/s. Graph v as a function of time t , measured in seconds.
- (ii) Let the distance travelled by the car be s metres from where it had stopped. On a separate diagram, graph s as a function of time t .



- (ii) The distance travelled is given by the area under the v/t curve.

$$\text{After 10 s, } s = \frac{1}{2} \times 10 \times 25 = 125 \text{ m}$$

$$\text{After 30 s, } s = 125 + (20 \times 25) = 625 \text{ m}$$

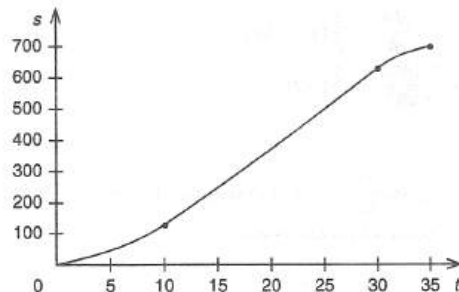
$$\text{After 35 s, } s = 625 + \frac{1}{2}(5 + 25) \cdot 5 = 625 + 75 = 700 \text{ m}$$

The shape of the s/t curve is determined by the v/t relationship:

From $t = 0$ s to $t = 10$ s, the v/t relationship is linear, $\therefore$ the s/t relationship is parabolic.

From $t = 10$ s to $t = 30$ s, the v/t relationship is constant, $\therefore$ the s/t relationship is linear.

From $t = 30$ s to $t = 35$ s, the v/t relationship is linear, $\therefore$ the s/t relationship is parabolic.



Q 4

1994

The number of unemployed people u at time t was studied over a period of time. At the start of this period, the number of unemployed was 800 000.

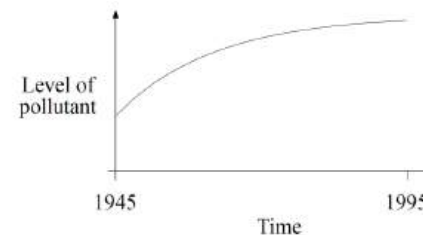
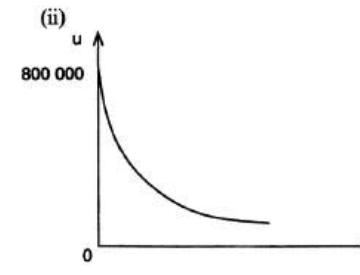
- (i) Throughout the period, $\frac{du}{dt} < 0$.

What does this say about the number of unemployed during the period?

- (ii) It is also observed that, throughout the period, $\frac{d^2u}{dt^2} > 0$.

Sketch the graph of u against t .

- (a) (i) The number of unemployed decreased during the period.



Q 5

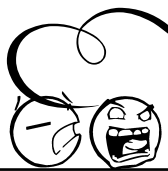
1995

The graph shows the levels of a pollutant in the atmosphere over the past 50 years.

Describe briefly how the level of this pollutant has changed over this period of time. Include mention of the rate of change.

- (i) The level of pollutant has risen from 1945 to 1995, but the rate of increase has decreased to almost zero at 1995.





INTRODUCTION TO CALCULUS BOS

PART H: RATES OF CHANGE



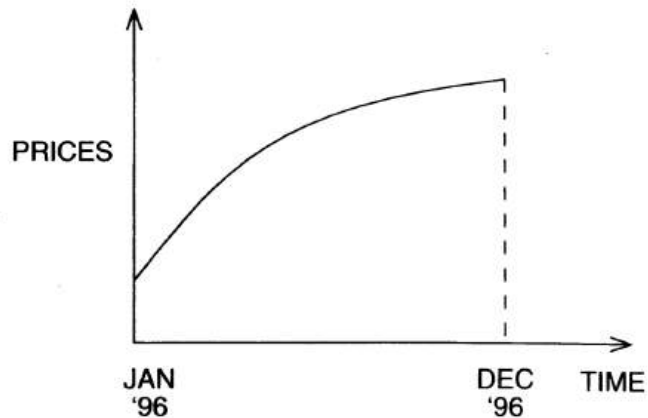
Unit

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Q 6

1997

The rate of inflation measures the rate of change in prices. Between January 1996 and December 1996, prices were rising but the rate of inflation was falling. Draw a graph of prices as a function of time that fits this description.



Q 7

2002

2

A cooler, which is initially full, is drained so that at time t seconds the volume

of water V , in litres, is given by $V = 25\left(1 - \frac{t}{60}\right)^2$ for $0 \leq t \leq 60$.

- How much water was initially in the cooler? 1
- After how many seconds was the cooler one-quarter full? 2
- At what rate was the water draining out when the cooler was one-quarter full? 2

(i) $V = 25\left(1 - \frac{t}{60}\right)^2$ for $0 \leq t \leq 60$

$$\begin{aligned} \text{At } t = 0, V &= 25\left(1 - \frac{0}{60}\right)^2 \\ &= 25 \times 1 \\ &= 25 \text{ litres.} \end{aligned}$$

$$= -\frac{5}{12} \text{ litres/second.}$$

(Negative value indicates water is draining out.)

(ii) $V = 25\left(1 - \frac{t}{60}\right)^2$

When $V = \frac{25}{4}$,

$$\frac{25}{4} = 25\left(1 - \frac{t}{60}\right)^2$$

$$\frac{1}{4} = \left(1 - \frac{t}{60}\right)^2$$

$$\pm \frac{1}{2} = 1 - \frac{t}{60}$$

$$\frac{t}{60} = 1 \pm \frac{1}{2}$$

$$\frac{t}{60} = 1\frac{1}{2} \text{ or } \frac{t}{60} = \frac{1}{2}$$

$$\therefore t = 90 \text{ or } t = 30$$

$$\therefore t = 30 \text{ seconds } 0 \leq t \leq 60.$$

(ii) Now $\frac{dV}{dt} = 2 \times 25\left(1 - \frac{t}{60}\right)^1 \cdot \frac{d}{dt}\left(1 - \frac{t}{60}\right)$

$$= 50\left(1 - \frac{t}{60}\right) \cdot -\frac{1}{60}$$

When $t = 30$,

$$\frac{dV}{dt} = 50\left(1 - \frac{30}{60}\right) \times -\frac{1}{60}$$

$$= 50\left(\frac{1}{2}\right)\left(-\frac{1}{60}\right)$$

$$= -\frac{50}{120}$$



INTRODUCTION TO CALCULUS BOS

PART h: Rates of Change

Q 8

2005

A tank initially holds 3600 litres of water. The water drains from the bottom of the tank. The tank takes 60 minutes to empty.

A mathematical model predicts that the volume, V litres, of water that will remain in the tank after t minutes is given by

$$V = 3600 \left(1 - \frac{t}{60}\right)^2, \text{ where } 0 \leq t \leq 60.$$

- | | |
|--|---|
| (i) What volume does the model predict will remain after ten minutes? | 1 |
| (ii) At what rate does the model predict that the water will drain from the tank after twenty minutes? | 2 |
| (iii) At what time does the model predict that the water will drain from the tank at its fastest rate? | 2 |

$$V = 3600 \left(1 - \frac{t}{60}\right)^2, \quad 0 \leq t \leq 60.$$

(i) When $t = 10$, $V = 3600 \left(1 - \frac{10}{60}\right)^2$
 $= 2500 \text{ L.}$

$\therefore$ After 10 minutes, 2500 L will remain in the tank.

(ii) Rate of draining $= \frac{dV}{dt}$.

$$\frac{dV}{dt} = 3600 \times 2 \left(1 - \frac{t}{60}\right) \times \left(-\frac{1}{60}\right)$$

$$= -120 \left(1 - \frac{t}{60}\right).$$

When $t = 20$, $\frac{dV}{dt} = -120 \left(1 - \frac{20}{60}\right)$
 $= -80.$

$\therefore$ After 20 minutes, water will drain out of the tank at a rate of 80 L/min.

- (iii) Fastest rate when $\frac{dV}{dt}$ is a maximum.

METHOD 1

Now $\frac{dV}{dt} = -120 \left(1 - \frac{t}{60}\right).$

(Note: $0 \leq t \leq 60$)

$\therefore$ Maximum draining rate when $t = 0$
 (ie. 120 L/min).

METHOD 2

